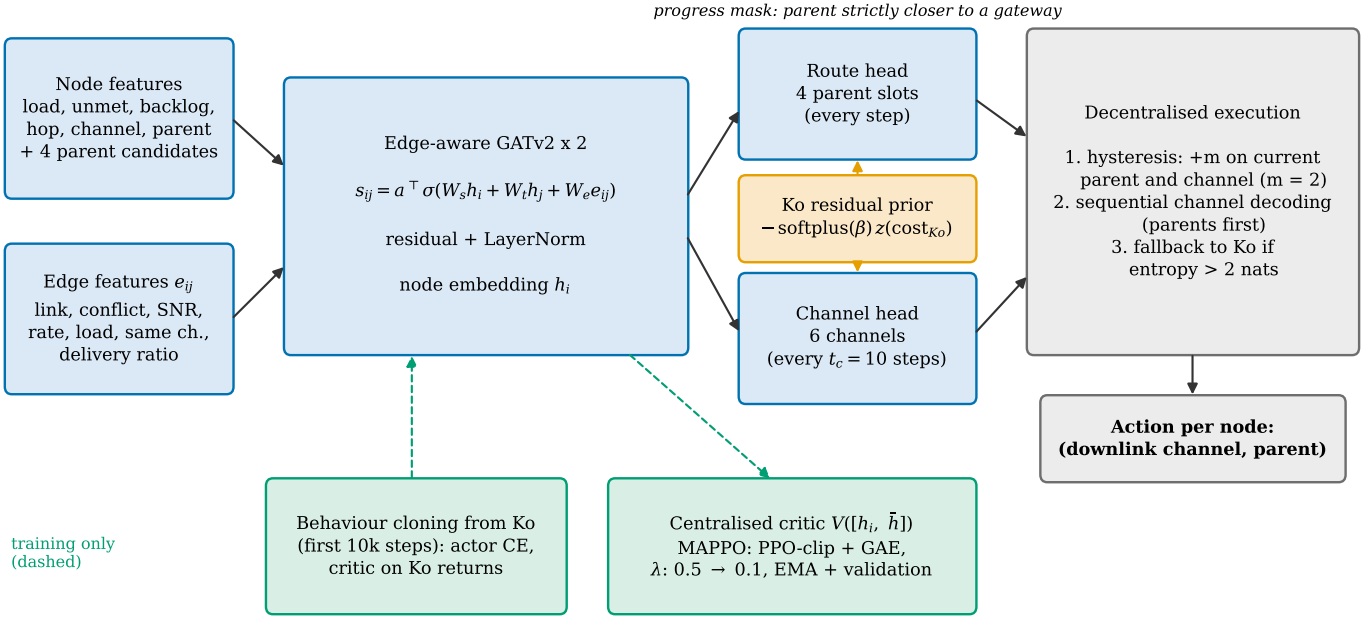
progress mask: parent strictly closer to a gateway
Node features
load, unmet, backlog,
hop, channel, parent
+ 4 parent candidates
Edge features e_{ij}
link, conflict, SNR,
rate, load, same ch.,
delivery ratio
Edge-aware GATv2 x 2
$s_{ij} = a^\top \sigma(W_s h_i + W_t h_j + W_e e_{ij})$
residual + LayerNorm
node embedding h_i
Route head
4 parent slots
(every step)
Ko residual prior
$-\mathrm{softplus}(\beta)\, z(\mathrm{cost}_{Ko})$
Channel head
6 channels
(every $t_c = 10$ steps)
Decentralised execution
1. hysteresis: +m on current
parent and channel (m = 2)
2. sequential channel decoding
(parents first)
3. fallback to Ko if
entropy > 2 nats
Action per node:
(downlink channel, parent)
training only
(dashed)
Behaviour cloning from Ko
(first 10k steps): actor CE,
critic on Ko returns
Centralised critic $V([h_i,\ \bar{h}])$
MAPPO: PPO-clip + GAE,
λ: 0.5 → 0.1, EMA + validation